Quantum Computing Unlocked

Quantum computing is poised to revolutionise industries by finding solutions to challenges once deemed intractable.

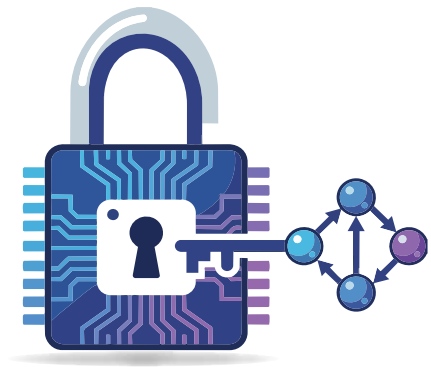
Quantum computing has moved beyond academic exploration into early commercial deployment, with active pilots across finance, healthcare, logistics, and cybersecurity. Enterprises are running proof-of-concepts for portfolio optimisation, molecular simulation, and quantum-safe encryption to prepare for the arrival of a cryptographically relevant quantum computer.

According to McKinsey's 2026 Quantum Technology Monitor, more than 300 global companies including Airbus, JPMorgan Chase, Boehringer Ingelheim, and Liberty Mutual are now actively collaborating with quantum technology vendors, and the technology could unlock as much as US$ 2.7 trillion in economic value across industries by 2035.

Classical computing vs quantum computing

Conventional silicon-based computers consist of millions of transistors that switch between 'on' and 'off', corresponding to the two possible values of bits — 0 and 1. Processing relies on basic logic gates AND, NOT, and the like, acting on individual bits or pairs of bits. Each bit's state can be measured and distinguished without disturbing the system.

Quantum computers, by contrast, employ quantum bits or qubits that can occupy a superposition of both 0 and 1 at the same time. A two-qubit device can represent four states simultaneously, a three-qubit system eight states, a four-qubit system sixteen states, and an n-qubit can register up to 2^n unique states in parallel. Qubits can also become entangled, which means the state of one qubit



is intrinsically linked to the state of another — a defining feature of quantum mechanics.

This inherent parallelism allows quantum machines to tackle specific classes of complex problems that are far beyond the reach of classical architectures.

Characteristics of a quantum computing platform (QCP)

Low level programming: Quantum computers are built on low level programming based on quantum logical gates to handle computational steps that execute in quantum processing units (QPUs). Examples of quantum logical gates are Pauli, Hadamard, CCNot, etc.

Heterogeneous: Technical specifications of the various quantum computing platforms (QCPs) are heterogeneous with respect to both the software and hardware in QPUs.

Remote software development and deployment: QCP vendors provide the quantum computing software development framework for leveraging quantum processors that are accessed

Some quantum facts

- Following the release of official PQC (post-quantum cryptography) standards by NIST (e.g., ML-KEM, ML-DSA, SLH-DSA), enterprises are actively executing migration strategies to protect legacy data systems against future quantum-enabled decryption threats.
- IBM, Google, Microsoft, Quantinuum, and IonQ have all hit logical-qubit milestones. Highlights include Microsoft/Quantinuum's 12 logical qubits (March 2026), IBM's 4,158-qubit Kookaburra system, and Google's proof of exponential error suppression via surface-code scaling.
- According to the Grand View Research report on quantum computing, the US government acquired equity worth US$ 2 billion in nine quantum firms. D-Wave Systems acquired Quantum Computing Inc. for US$ 550 million and Google/Nvidia backed QuEra's US$ 230+ million funding round.
- A Roots Analysis report says that Quantum Computing-as-a-Service (cloud-based access) captured approximately 40% share in 2026, since it removes the need for costly hardware investment.
- A BQPSim report says that, in 2026, AWS Braket, Google Quantum AI, IBM Quantum Network, and Microsoft Azure Quantum all expanded pay-as-you-go access to quantum processing units.
- According to Stellar Market Research, healthcare and pharmaceuticals are projected to see the highest CAGR (around 29%) in quantum computing among end-use segments through the mid-2030s. Growth will be driven by drug discovery, genome sequencing, and molecular simulation.